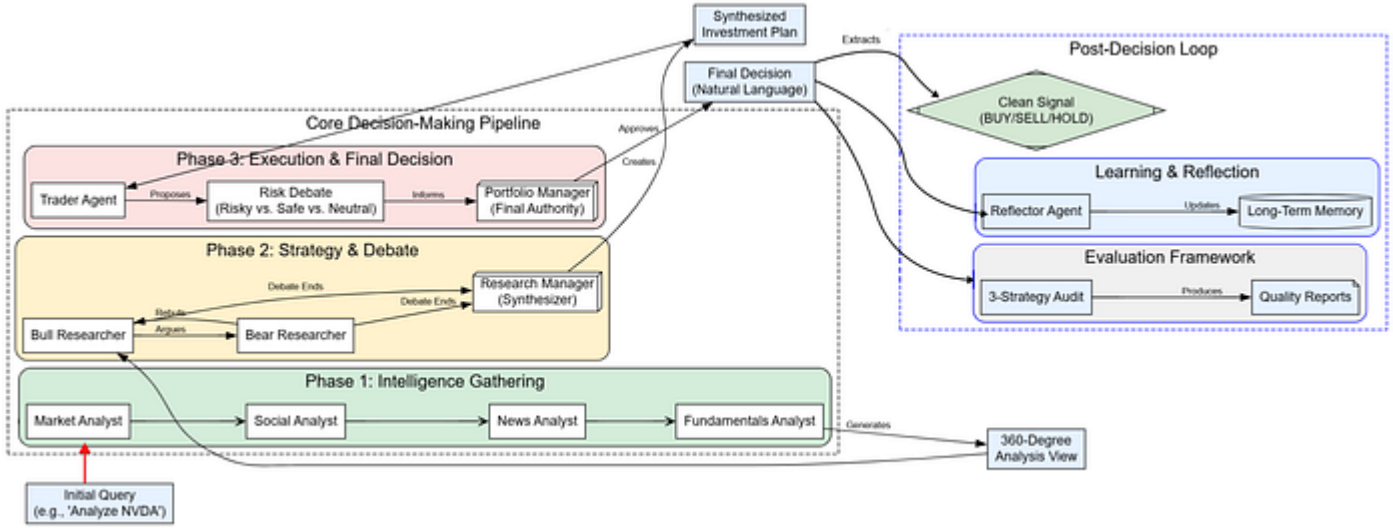

Building a Deep Thinking Trading System with Multi-Agentic Architecture

Analyst, Debate, Trader, Risk, Evaluation and more

Fareed Khan

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A Deep Thinking Trading system has many departments, each made up of sub-agents that use logical flows to make smart decisions. For example, an Analyst team gathers data from diverse sources, a Researcher team debates and analyzes this data to form a strategy, and the Execution team refines and approves the trade while working alongside portfolio management, other supporting sub-agents, and more.

There is a lot that happens under the hood, a typical flow works like this ...

1. **First, a team of specialized Analyst agents gathers 360-degree market intelligence,** collecting everything from technical data and news to social media sentiment and company fundamentals.
2. **Then, a Bull and a Bear agent engage in an adversarial debate** to stress-test the findings, which a Research Manager synthesizes into a single, balanced investment strategy.
3. **Next, a Trader agent translates this strategy into a concrete, actionable proposal,** which is immediately scrutinized by a multi-perspective Risk Management team (Risky, Safe, and Neutral).
4. **The final, binding decision is made by a Portfolio Manager agent,** who weighs the Trader's plan against the risk debate to give the final approval.

and auditing.

6. **Finally, the entire process creates a feedback loop.** Agents reflect on the trade's outcome to generate new lessons, which are stored in their long-term memory to continuously improve future performance.

> In this blog, we will code and visualize an advanced agent-based trading system where Analysts, Researchers, Traders, Risk Managers, and a Portfolio Manager work together to execute smart trades.

If you are new to finance, Please take a look at these two articles to understand the fundamentals:

- **9 Must-Know Financial Terms for Beginners from Holborn Assets** Covers essential finance concepts like **assets**, **liabilities**, **net worth**, **cash flow**, **interest rate**, **diversification**, **inflation**, and more perfect for building a solid financial vocabulary. holbornassets.sa
- **Stock Market Basics: What Beginner Investors Should Know NerdWallet**, Explains how the stock market works, the role of brokers, exchanges, indexes (like the NYSE, Nasdaq, S&P 500), and how to get started with a brokerage account.

GitHub - FareedKhan-dev/multi-agent-trading-system: Implementation of Deep Thinking Trading System

Implementation of Deep Thinking Trading System. Contribute to FareedKhan-dev/multi-agent-trading-system development by...

github.com

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Setting up the Environment, LLMs, and LangSmith Tracing

First, we should keep our API keys safe and set up tracing with LangSmith. Tracing is important because it lets us see what's happening inside our multi-agent system step by step. This makes it much easier to debug problems and understand how the agents are working. If we want a reliable system in production, tracing isn't optional, it's mandatory.

Copy

```
# First, ensure you have the necessary libraries installed
# !pip install -U langchain langgraph langchain_openai tavily-python yfinance t

import os
from getpass import getpass

# Define a helper function to set environment variables securely.
def _set_env(var: str):
```

```
        os.environ[var] = getpass(f"Enter your {var}: ")

# Set API keys for the services we'll use.
_set_env("OPENAI_API_KEY")
_set_env("FINNHUB_API_KEY")
_set_env("TAVILY_API_KEY")
_set_env("LANGSMITH_API_KEY")

# Enable LangSmith tracing for full observability of our agentic system.
os.environ["LANGSMITH_TRACING"] = "true"

# Define the project name for LangSmith to organize traces.
os.environ["LANGSMITH_PROJECT"] = "Deep-Trading-System"
```

We're setting API keys for **OpenAI**, **Finnhub**, and **Tavily**.

- OpenAI to runs our LLMs (you can also test with free credits from open-source LLM providers like **Together AI**).
- Finnhub for real-time stock market data (free tier gives limited API calls).
- Tavily for web search & news (free tier available).
- LangSmith for tracing and observability (free tier lets you track and debug agents easily).

to isolate and analyze this specific execution.

To make our system modular, we will use a central configuration dictionary. This acts as our control panel, allowing us to easily experiment with different models or parameters without changing the core agent logic.

Copy

```
from pprint import pprint

# Define our central configuration for this notebook run.
config = {
    "results_dir": "./results",
    # LLM settings specify which models to use for different cognitive tasks.
    "llm_provider": "openai",
    "deep_think_llm": "gpt-4o",       # A powerful model for complex reasoning
    "quick_think_llm": "gpt-4o-mini", # A fast, cheaper model for data processi
    "backend_url": "https://api.openai.com/v1",
    # Debate and discussion settings control the flow of collaborative agents.
    "max_debate_rounds": 2,          # The Bull vs. Bear debate will have 2 rou
    "max_risk_discuss_rounds": 1,    # The Risk team has 1 round of debate.
    "max_recur_limit": 100,          # Safety limit for agent loops.
    # Tool settings control data fetching behavior.
    "online_tools": True,            # Use live APIs; set to False to use cache
    "data_cache_dir": "./data_cache" # Directory for caching online data.
}
```

```
print("Configuration dictionary created.")
pprint(config)
```

These parameters will make sense to you more later in the blog but let's break down a few key parameters here. The choice of two different models (`deep_think_llm` and `quick_think_llm`) is a deliberate architectural decision to optimize for both cost and performance.**deep_think_llm (`gpt-4o`)** → handles complex reasoning and final decisions.

- quick_think_llm (`gpt-4o-mini`) for faster, cheaper for routine tasks.
- max_debate_rounds is for controls how many rounds the Bull vs. Bear agents argue.
- max_risk_discuss_rounds is for how long the Risk team debates.
- online_tools / data_cache_dir let us switch between live API calls and cached data.

With our configuration defined, we can now initialize the LLMs that will serve as the cognitive engines for our agents.

```
Copy
```

```
# Initialize the powerful LLM for high-stakes reasoning tasks.
deep_thinking_llm = ChatOpenAI(
    model=config["deep_think_llm"],
    base_url=config["backend_url"],
    temperature=0.1
)
# Initialize the faster, cost-effective LLM for routine data processing.
quick_thinking_llm = ChatOpenAI(
    model=config["quick_think_llm"],
    base_url=config["backend_url"],
    temperature=0.1
)
```

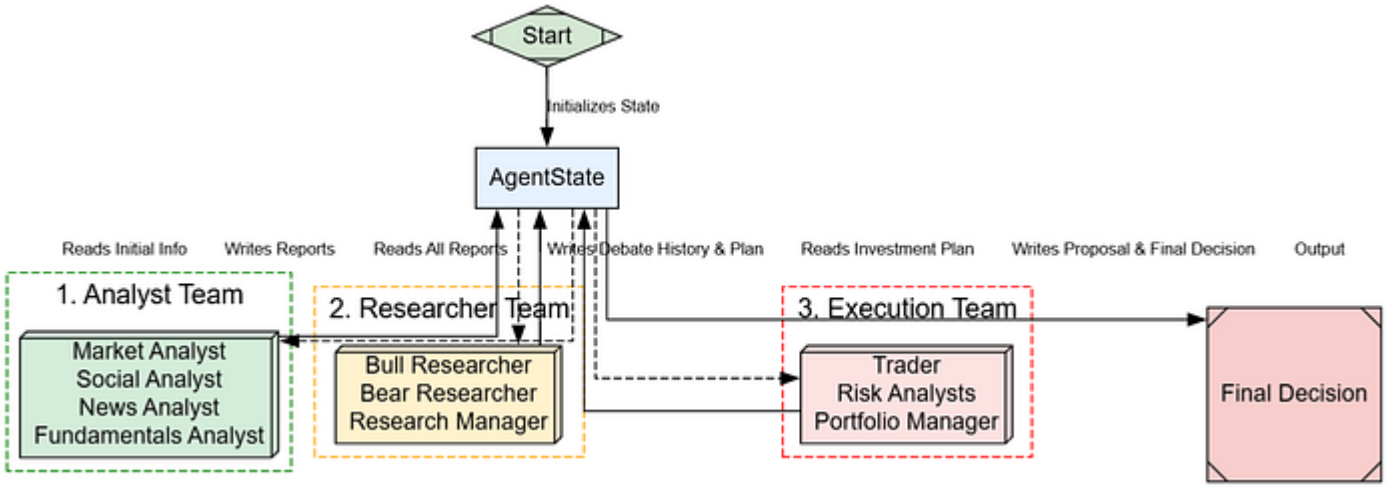
We have now instantiated our two LLMs. Note the `temperature` parameter is set to `0.1`.

> For financial analysis, we want responses that are deterministic and fact-based, not highly creative.

A low temperature encourages the model to stick to the most likely, grounded outputs.

Designing the Shared Memory

In LangGraph, the state is a central object that is passed between all nodes. As each agent completes its task, it reads from and writes to this state.



Agent State Purpose (Created by Fareed Khan)

We will define the data structures for this global memory using Python `TypedDict`. This gives us a strongly-typed, predictable structure, which is important for managing complexity. We will start by defining the sub-states for our two debate teams.

```
from typing import Annotated, Sequence, List
from typing_extensions import TypedDict
from langgraph.graph import MessagesState

# State for the researcher team's debate, acting as a dedicated scratchpad.
class InvestDebateState(TypedDict):
    bull_history: str      # Stores arguments made by the Bull agent.
    bear_history: str      # Stores arguments made by the Bear agent.
    history: str           # The full transcript of the debate.
    current_response: str  # The most recent argument made.
    judge_decision: str    # The manager's final decision.
    count: int             # A counter to track the number of debate rounds.

# State for the risk management team's debate.
class RiskDebateState(TypedDict):
    risky_history: str     # History of the aggressive risk-taker.
    safe_history: str      # History of the conservative agent.
    neutral_history: str   # History of the balanced agent.
    history: str           # Full transcript of the risk discussion.
    latest_speaker: str    # Tracks the last agent to speak.
    current_risky_response: str
    current_safe_response: str
    current_neutral_response: str
    judge_decision: str    # The portfolio manager's final decision.
    count: int             # Counter for risk discussion rounds.
```

`InvestDebateState` and `RiskDebateState` act as dedicated scratchpads. We are designing this pattern as it keeps the state organized and prevents different

for our graph conditional logic to know when to end the debate.

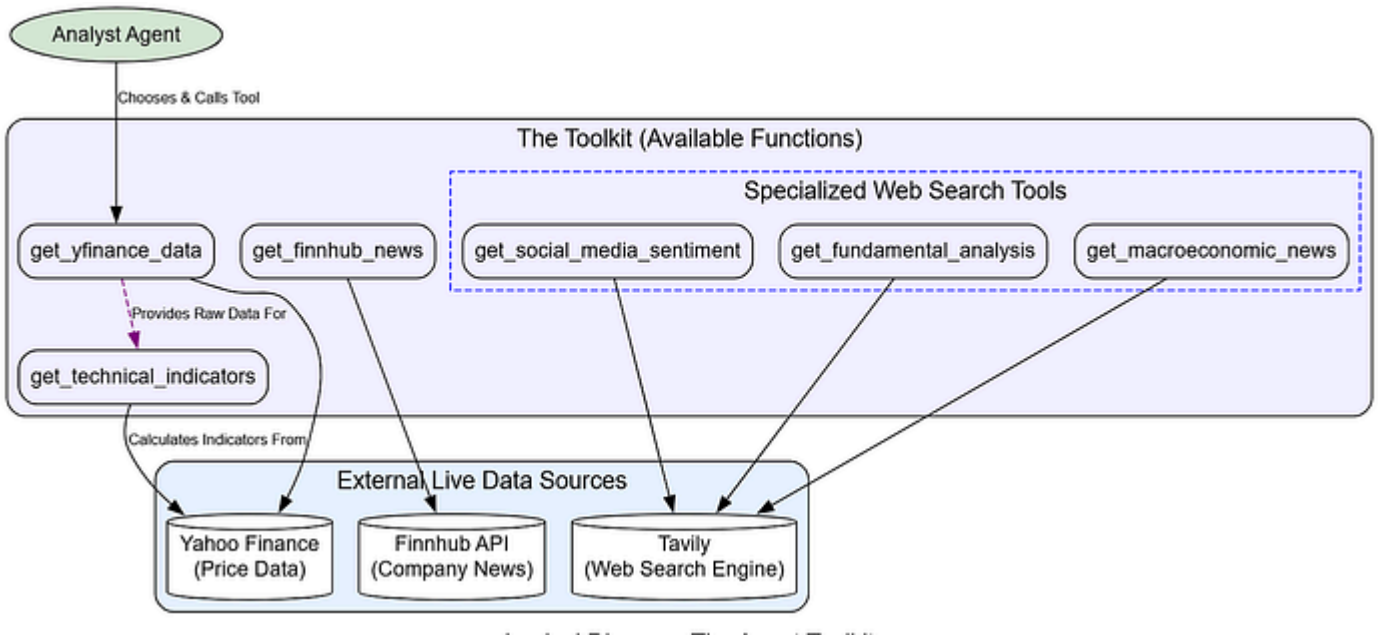
Now, we define the main `AgentState` that incorporates these sub-states.

```
Copy

# The main state that will be passed through the entire graph.
# It inherits from MessagesState to include a 'messages' field for chat history
class AgentState(MessagesState):
    company_of_interest: str          # The stock ticker we are analyzing.
    trade_date: str                   # The date for the analysis.
    sender: str                       # Tracks which agent last modified the st
    # Each analyst will populate its own report field.
    market_report: str
    sentiment_report: str
    news_report: str
    fundamentals_report: str
    # Nested states for the debates.
    investment_debate_state: InvestDebateState
    investment_plan: str              # The plan from the Research Manager.
    trader_investment_plan: str       # The actionable plan from the Trader.
    risk_debate_state: RiskDebateState
    final_trade_decision: str         # The final decision from the Portfolio M
```

debate states we just defined. This structured approach allows us to trace the flow of information from the initial raw data all the way to the `final_trade_decision`.

Building the Agent Toolkit with Live Data

An agent is only as good as its tools. So we need a `Toolkit` where we define all the functions our agents can use to interact with the outside world. This is what grounds their reasoning in real, up-to-the-minute data.



Each function is decorated with `@tool` from LangChain, along with `Annotated` type hints, providing a schema that the LLM uses to understand the tool's purpose and inputs. Let's start with a tool to fetch raw historical price data.

Copy

```
import yfinance as yf
from langchain_core.tools import tool

@tool
def get_yfinance_data(
    symbol: Annotated[str, "ticker symbol of the company"],
    start_date: Annotated[str, "Start date in yyyy-mm-dd format"],
    end_date: Annotated[str, "End date in yyyy-mm-dd format"],
) -> str:
    """Retrieve the stock price data for a given ticker symbol from Yahoo Finar
    try:
        ticker = yf.Ticker(symbol.upper())
        data = ticker.history(start=start_date, end=end_date)
        if data.empty:
            return f"No data found for symbol '{symbol}' between {start_date} a
        return data.to_csv()
    except Exception as e:
        return f"Error fetching Yahoo Finance data: {e}"
```

parameter (e.g., that `symbol` should be a **"ticker symbol"**).

We return the data as a `.to_csv()` string, a simple format that LLMs are very effective at parsing. Next, we need a tool to derive common technical indicators from that raw data.

Copy

```
from stockstats import wrap as stockstats_wrap

@tool
def get_technical_indicators(
    symbol: Annotated[str, "ticker symbol of the company"],
    start_date: Annotated[str, "Start date in yyyy-mm-dd format"],
    end_date: Annotated[str, "End date in yyyy-mm-dd format"],
) -> str:
    """Retrieve key technical indicators for a stock using stockstats library.'
    try:
        df = yf.download(symbol, start=start_date, end=end_date, progress=False
        if df.empty:
            return "No data to calculate indicators."
        stock_df = stockstats_wrap(df)
        indicators = stock_df[['macd', 'rsi_14', 'boll', 'boll_ub', 'boll_lb',
        return indicators.tail().to_csv()
    except Exception as e:
        return f"Error calculating stockstats indicators: {e}"
```

This is an important practical consideration to keep the context passed to the LLM concise and relevant, avoiding unnecessary tokens and cost.

Now for company-specific news, using the Finnhub API.

```
Copy

import finnhub

@tool
def get_finnhub_news(ticker: str, start_date: str, end_date: str) -> str:
    """Get company news from Finnhub within a date range."""
    try:
        finnhub_client = finnhub.Client(api_key=os.environ["FINNHUB_API_KEY"])
        news_list = finnhub_client.company_news(ticker, _from=start_date, to=er
        news_items = []
        for news in news_list[:5]: # Limit to 5 results
            news_items.append(f"Headline: {news['headline']}\nSummary: {news['s
        return "\n\n".join(news_items) if news_items else "No Finnhub news four
    except Exception as e:
        return f"Error fetching Finnhub news: {e}"
```

The tools above provide structured data. For less tangible factors like market buzz, we need a general web search tool. We will use **Tavily**. We will initialize it once for efficiency.

```
from langchain_community.tools.tavily_search import TavilySearchResults

# Initialize the Tavily search tool once. We can reuse this instance for multip
tavily_tool = TavilySearchResults(max_results=3)
```

Now we create three specialized search tools. This is a critical design choice. While they all use the same Tavily engine, providing the LLM with distinct tools like `get_social_media_sentiment` versus a generic `search_web` simplifies its decision-making process. It's easier for the LLM to choose the right tool for the job when its purpose is narrow and clearly defined.

Copy

```
@tool
def get_social_media_sentiment(ticker: str, trade_date: str) -> str:
    """Performs a live web search for social media sentiment regarding a stock.
    query = f"social media sentiment and discussions for {ticker} stock around
    return tavily_tool.invoke({"query": query})

@tool
def get_fundamental_analysis(ticker: str, trade_date: str) -> str:
    """Performs a live web search for recent fundamental analysis of a stock."'
    query = f"fundamental analysis and key financial metrics for {ticker} stock
    return tavily_tool.invoke({"query": query})
@tool
def get_macroeconomic_news(trade_date: str) -> str:
```

```
    return tavily_tool.invoke({"query": query})
```

Finally, we'll aggregate all these functions into a single `Toolkit` class for clean, organized access.

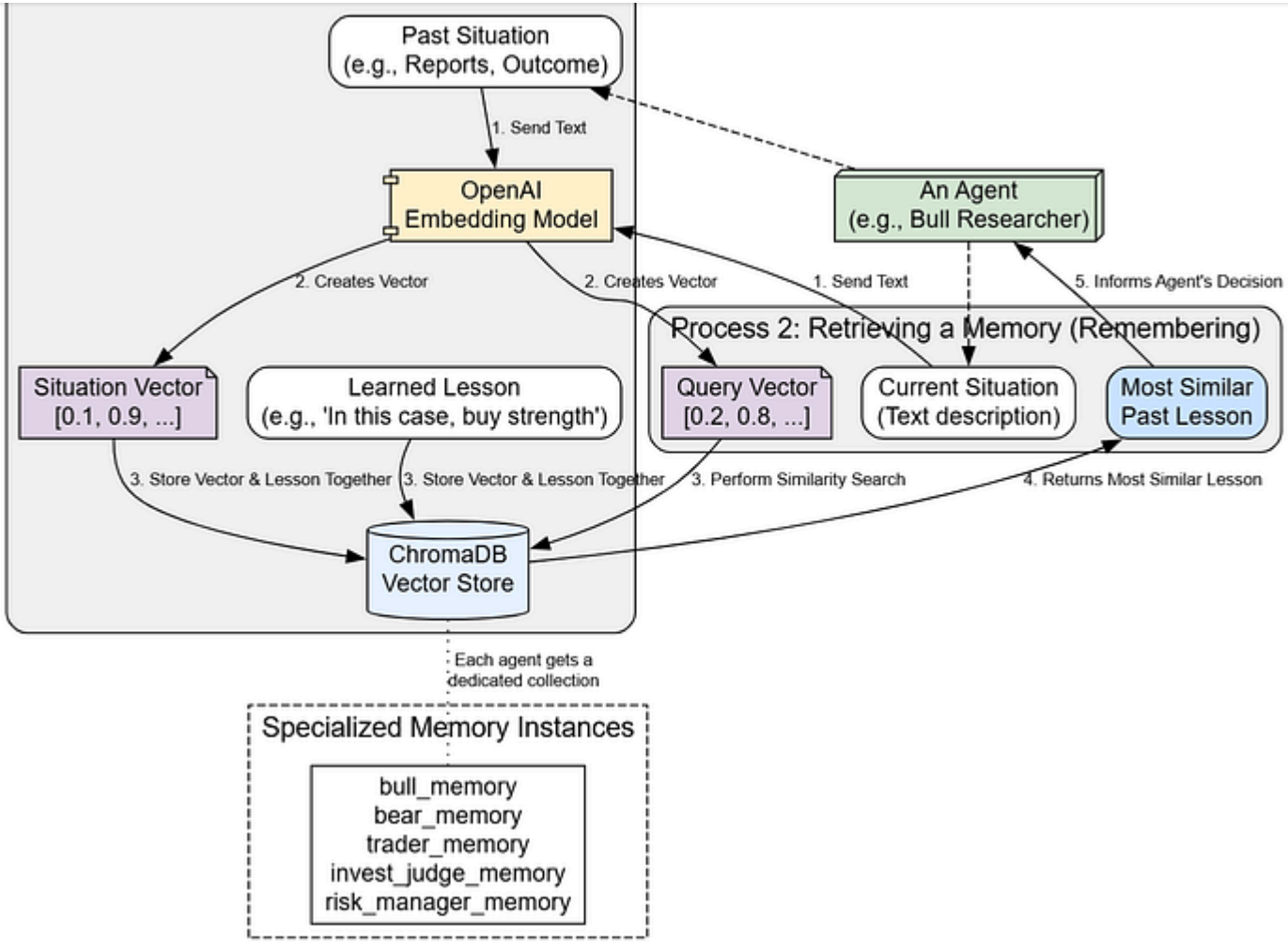
Copy

```
# The Toolkit class aggregates all defined tools into a single, convenient obje
class Toolkit:
    def __init__(self, config):
        self.config = config
        self.get_yfinance_data = get_yfinance_data
        self.get_technical_indicators = get_technical_indicators
        self.get_finnhub_news = get_finnhub_news
        self.get_social_media_sentiment = get_social_media_sentiment
        self.get_fundamental_analysis = get_fundamental_analysis
        self.get_macroeconomic_news = get_macroeconomic_news

# Instantiate the Toolkit, making all tools available through this single objec
toolkit = Toolkit(config)
print(f"Toolkit class defined and instantiated with live data tools.")
```

Our foundation for agent action is now complete. We have a comprehensive suite of tools that allows our system to gather a rich, multi-faceted view of any financial asset.

Implementing Long-Term Memory for Continuous Learning

For our agents to improve over time, they need long-term memory. To enable true learning, agents must be able to reflect on past decisions and retrieve those lessons when faced with similar situations in the future.



Long term memory flow (Created by Fareed Khan)

The `FinancialSituationMemory` class is the core of this learning loop. It uses a `ChromaDB` vector store to save textual summaries of past situations and the lessons learned from them.

```
import chromadb
from openai import OpenAI

# The FinancialSituationMemory class provides long-term memory
# for storing and retrieving financial situations + recommendations.
class FinancialSituationMemory:
    def __init__(self, name, config):
        # Use OpenAI's small embedding model for vectorizing text
        self.embedding_model = "text-embedding-3-small"

        # Initialize OpenAI client (pointing to your configured backend)
        self.client = OpenAI(base_url=config["backend_url"])

        # Create a ChromaDB client (with reset allowed for testing)
        self.chroma_client = chromadb.Client(chromadb.config.Settings(allow_res

        # Create a collection (like a table) to store situations + advice
        self.situation_collection = self.chroma_client.create_collection(name=r

    def get_embedding(self, text):
        # Generate an embedding (vector) for the given text
        response = self.client.embeddings.create(model=self.embedding_model, in
        return response.data[0].embedding

    def add_situations(self, situations_and_advice):
        # Add new situations and recommendations to memory
        if not situations_and_advice:
            return
```

```
        ids = [str(offset + i) for i, _ in enumerate(situations_and_advice)]

        # Separate situations and their corresponding advice
        situations = [s for s, r in situations_and_advice]
        recommendations = [r for s, r in situations_and_advice]

        # Generate embeddings for all situations
        embeddings = [self.get_embedding(s) for s in situations]

        # Store everything in Chroma (vector DB)
        self.situation_collection.add(
            documents=situations,
            metadatas=[{"recommendation": rec} for rec in recommendations],
            embeddings=embeddings,
            ids=ids,
        )

    def get_memories(self, current_situation, n_matches=1):
        # Retrieve the most similar past situations for a given query
        if self.situation_collection.count() == 0:
            return []

        # Embed the new/current situation
        query_embedding = self.get_embedding(current_situation)

        # Query the collection for similar embeddings
        results = self.situation_collection.query(
            query_embeddings=[query_embedding],
            n_results=min(n_matches, self.situation_collection.count()),
```

```
        # Return extracted recommendations from the matches
        return [{'recommendation': meta['recommendation']} for meta in results
```

Let's break down the key methods.

1. The `__init__` method sets up a unique ChromaDB `collection` for each agent, giving them their own private memory space.
2. The `add_situations` method is where learning happens: it takes a situation (the context) and a lesson, creates a vector embedding of the situation, and stores both in the database.
3. The `metadatas` parameter is used to store the actual lesson text, while the `documents` store the context.
4. Finally, `get_memories` is the retrieval step. It takes the `current_situation`, creates a `query_embedding`, and performs a similarity search to find the most relevant lessons from the past.

Now, we will create a dedicated memory instance for each learning agent.

```
                                    Copy

# Create a dedicated memory instance for each agent that learns.
bull_memory = FinancialSituationMemory("bull_memory", config)
```

```
invest_judge_memory = FinancialSituationMemory("invest_judge_memory", config)
risk_manager_memory = FinancialSituationMemory("risk_manager_memory", config)
```

We have now created five separate memory instances. This specialization is important because a lesson learned by the Bull agent (e.g., "In a strong uptrend, valuation concerns are less important") might not be relevant for the more conservative Safe Risk Analyst.

> By giving them separate memories, we ensure the lessons they retrieve are specific to their roles.

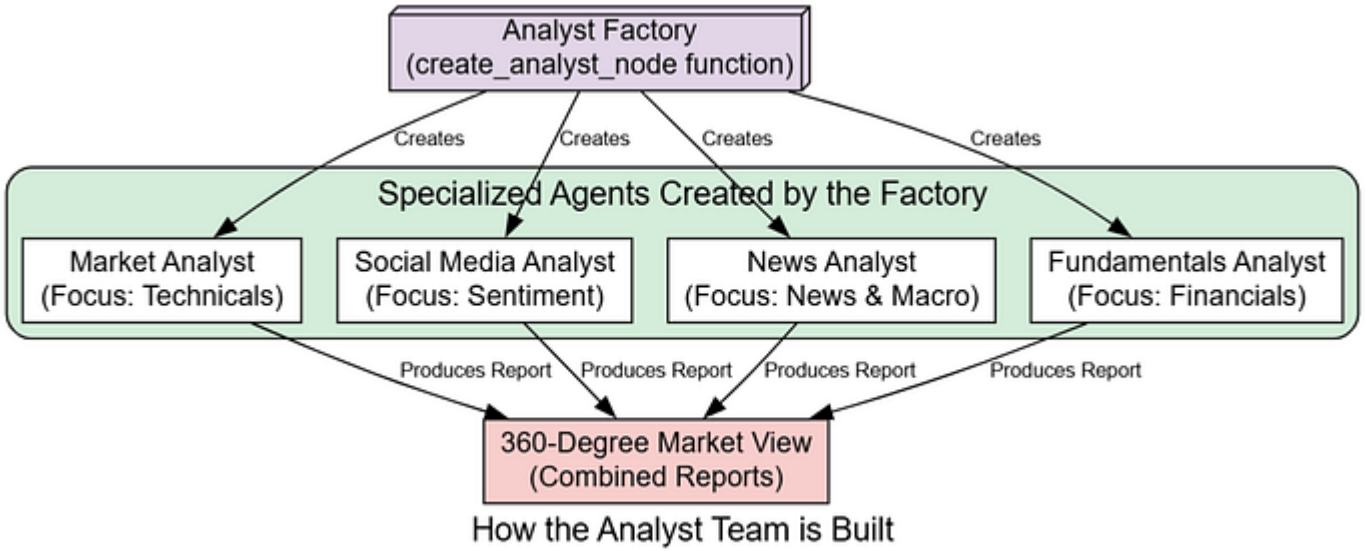
With our environment, state, tools, and memory all defined, the foundational layer of our system is complete. We are now ready to start building the agents themselves.

Deploying the Analyst Team for 360-Degree Market Intelligence

With our foundation in place, it's time to introduce the first team of agents, the **Analysts**. This team is the intelligence-gathering arm of our firm.

angles.

We will therefore create **four specialized analysts**, each responsible for a unique domain.



Analyst Factory (Created by Fareed Khan)

Their collective goal is to produce a comprehensive, 360-degree view of the stock and its market environment, populating our `AgentState` with the raw intelligence needed for the next stage of strategic analysis.

from a common template.

Copy

```
from langchain_core.prompts import ChatPromptTemplate, MessagesPlaceholder

# This function is a factory that creates a LangGraph node for a specific type
def create_analyst_node(llm, toolkit, system_message, tools, output_field):
    """
    Creates a node for an analyst agent.
    Args:
        llm: The language model instance to be used by the agent.
        toolkit: The collection of tools available to the agent.
        system_message: The specific instructions defining the agent's role and
        tools: A list of specific tools from the toolkit that this agent is al
        output_field: The key in the AgentState where this agent's final report
    """
    # Define the prompt template for the analyst agent.
    prompt = ChatPromptTemplate.from_messages([
        ("system",
         "You are a helpful AI assistant, collaborating with other assistants."
         " Use the provided tools to progress towards answering the question."
         " If you are unable to fully answer, that's OK; another assistant with
         " will help where you left off. Execute what you can to make progress.
         " You have access to the following tools: {tool_names}.\n{system_messa
         " For your reference, the current date is {current_date}. The company
        # MessagesPlaceholder allows us to pass in the conversation history.
        MessagesPlaceholder(variable_name="messages"),
```

```
    prompt = prompt.partial(system_message=system_message)
    prompt = prompt.partial(tool_names=", ".join([tool.name for tool in tools])
    # Bind the specified tools to the LLM. This tells the LLM which functions i
    chain = prompt | llm.bind_tools(tools)
    # This is the actual function that will be executed as a node in the graph.
    def analyst_node(state):
        # Fill in the final pieces of the prompt with data from the current sta
        prompt_with_data = prompt.partial(current_date=state["trade_date"], tic
        # Invoke the chain with the current messages from the state.
        result = prompt_with_data.invoke(state["messages"])
        report = ""
        # If the LLM didn't call a tool, it means it has generated the final re
        if not result.tool_calls:
            report = result.content
        # Return the LLM's response and the final report to update the state.
        return {"messages": [result], output_field: report}
    return analyst_node
```

Let's break down this factory function. `create_analyst_node` is a higher-order function that returns another function (`analyst_node`), which will serve as our actual node in the LangGraph workflow.

- The `system_message` parameter is crucial; it's where we inject the unique "personality" and goals for each analyst.
- The `tools` parameter enforces the division of labor—the Market Analyst can't access social media tools, and vice versa.

which to call.

Now that we have our factory, we can create our first specialist: the **Market Analyst**. Its job is purely quantitative, focusing on price action and technical indicators. We will equip it only with the tools necessary for this task.

```
Copy

# Market Analyst: Focuses on technical indicators and price action.
market_analyst_system_message = "You are a trading assistant specialized in ana
market_analyst_node = create_analyst_node(
    quick_thinking_llm,
    toolkit,
    market_analyst_system_message,
    [toolkit.get_yfinance_data, toolkit.get_technical_indicators],
    "market_report"
)
```

In this block, we define the `market_analyst_system_message` to give the agent a clear persona and objective. We then call our factory, passing in the `quick_thinking_llm`, the `toolkit`, this message, and a specific list of tools, `get_yfinance_data` and `get_technical_indicators`. Finally, we specify that its output should be saved to the `market_report` field in our `AgentState`.

```
Copy

# Social Media Analyst: Gauges public sentiment.
social_analyst_system_message = "You are a social media analyst. Your job is to
social_analyst_node = create_analyst_node(
    quick_thinking_llm,
    toolkit,
    social_analyst_system_message,
    [toolkit.get_social_media_sentiment],
    "sentiment_report"
)
```

Here, we've defined the agent responsible for sentiment analysis. Notice that its tool list is very narrow, containing only `get_social_media_sentiment` .

This specialization is a key principle of multi-agent design, ensuring each agent focuses on its core competency.

Our third specialist is the **News Analyst**, responsible for providing both company-specific and macroeconomic context.

```
Copy
```

```
news_analyst_node = create_analyst_node(
    quick_thinking_llm,
    toolkit,
    news_analyst_system_message,
    [toolkit.get_finnhub_news, toolkit.get_macroeconomic_news],
    "news_report"
)
```

The News Analyst is given two tools. This is a deliberate choice: `get_finnhub_news` provides micro-level, company-specific information, while `get_macroeconomic_news` provides the macro-level context. A comprehensive news analysis requires both perspectives.

Finally, our **Fundamentals Analyst** will investigate the company financial health.

Copy

```
# Fundamentals Analyst: Dives into the company's financial health.
fundamentals_analyst_system_message = "You are a researcher analyzing fundament
fundamentals_analyst_node = create_analyst_node(
    quick_thinking_llm,
    toolkit,
    fundamentals_analyst_system_message,
    [toolkit.get_fundamental_analysis],
```

It's important to understand *how* these agents work. They use a pattern called **ReAct (Reasoning and Acting)**. This isn't a single LLM call, it's a loop:

1. **Reason:** The LLM receives the prompt and decides if it needs a tool.
2. **Act:** If so, it generates a `tool_call` (e.g., `toolkit.get_yfinance_data(...)`).
3. **Observe:** Our graph executes this tool, and the result (the data) is passed back to the LLM.
4. **Repeat:** The LLM now has new information and decides its next step — either calling another tool or generating the final report.

This loop allows the agents to perform complex, multi-step data gathering tasks autonomously. To manage this loop, we need a helper function.

Copy

```
from langgraph.prebuilt import ToolNode, tools_condition
from langchain_core.messages import HumanMessage
import datetime
from rich.console import Console
from rich.markdown import Markdown

# Initialize a console for rich printing.
```

```
def run_analyst(analyst_node, initial_state):
    state = initial_state
    # Get all available tools from our toolkit instance.
    all_tools_in_toolkit = [getattr(toolkit, name) for name in dir(toolkit) if
    # The ToolNode is a special LangGraph node that executes tool calls.
    tool_node = ToolNode(all_tools_in_toolkit)
    # The ReAct loop can have up to 5 steps of reasoning and tool calls.
    for _ in range(5):
        result = analyst_node(state)
        # The tools_condition checks if the LLM's last message was a tool call.
        if tools_condition(result) == "tools":
            # If so, execute the tools and update the state.
            state = tool_node.invoke(result)
        else:
            # If not, the agent is done, so we break the loop.
            state = result
            break
    return state
```

The `run_analyst` function orchestrates the ReAct loop for a single agent. The `tools_condition` function is a LangGraph utility that inspects the agent's last message. If it contains a `tool_calls` attribute, it returns "tools", directing the flow to the `tool_node` for execution. Otherwise, the agent has finished its work.

Now, let's set our initial state and execute the first analyst.

```
TICKER = "NVDA"
# Use a recent date for live data fetching
TRADE_DATE = (datetime.date.today() - datetime.timedelta(days=2)).strftime('%Y-

# Define the initial state for the graph run.
initial_state = AgentState(
    messages=[HumanMessage(content=f"Analyze {TICKER} for trading on {TRADE_DAT
    company_of_interest=TICKER,
    trade_date=TRADE_DATE,
    # Initialize the debate states with default empty values.
    investment_debate_state=InvestDebateState({'history': '', 'current_response
    risk_debate_state=RiskDebateState({'history': '', 'latest_speaker': '', 'cu
)
# Run Market Analyst
print("Running Market Analyst...")
market_analyst_result = run_analyst(market_analyst_node, initial_state)
initial_state['market_report'] = market_analyst_result.get('market_report', 'Fa
console.print("----- Market Analyst Report -----")
console.print(Markdown(initial_state['market_report']))
```

We are targeting NVIDIA stocks for now, First we need to observe the market by first running the marketing analyst, this is what we get.

```
                                    Copy

# Running Market Analyst...
----- Market Analyst Report -----
```

```
| Indicator      | Signal       | Insight
|----------------|--------------|-----------------------------------------
| SMAs (50, 200) | Bullish      | Confirms a strong, sustained uptrend.
| MACD           | Bullish      | Positive momentum is currently in control.
| RSI (14)       | Strong       | Indicates strong buying pressure, but not yet
| Bollinger Bands| Expanding    | Suggests increasing volatility and potential
```

The Market Analyst has successfully executed its task. By inspecting the LangSmith trace, we would see a two-step process:

1. first, it called `get_yfinance_data` to fetch prices, then it used that data to call `get_technical_indicators`.
2. The final report is a clear, structured summary with a table, which is exactly what its prompt requested. This report is now stored in `initial_state['market_report']`.

The Market Analyst has given us a strong quantitative signal. Now, let's see if the public mood aligns with the charts.

Copy

```
# Run Social Media Analyst
print("\nRunning Social Media Analyst...")
social_analyst_result = run_analyst(social_analyst_node, initial_state)
initial_state['sentiment_report'] = social_analyst_result.get('sentiment_report
```

```
Running Social Media Analyst...
----- Social Media Analyst Report -----
Social media sentiment for NVDA is overwhelmingly bullish. Platforms like X (fo

| Platform      | Sentiment    | Key Themes
|---------------|--------------|-------------------------------------------
| X (Twitter)   | Very Bullish | AI Dominance, Analyst Upgrades, Product Hype
| Reddit        | Very Bullish | HODL mentality, Earnings Predictions, Memes
```

The Social Media Analyst is confirming the bullish thesis. Its report, generated from a live Tavily web search, captures the qualitative "hype" around the stock. This unstructured data is now distilled into a structured report and added to our `AgentState`.

But now we need to verify this confirmation by gathering the news context.

Copy

```
# Run News Analyst
print("\nRunning News Analyst...")
news_analyst_result = run_analyst(news_analyst_node, initial_state)
initial_state['news_report'] = news_analyst_result.get('news_report', 'Failed t
console.print("----- News Analyst Report -----")
console.print(Markdown(initial_state['news_report']))
Running News Analyst...
----- News Analyst Report -----
```

```
| News Category    | Impact     | Summary
|------------------|------------|-----------------------------------------
| Company-Specific | Positive   | New product announcements and strategic partne
| Macroeconomic    | Neutral+   | Stable inflation and interest rate outlook pro
```

The News Analyst report adds another layer, confirming that both company-specific and macroeconomic news are supportive. The agent correctly used both of its assigned tools to build this comprehensive picture.

Finally, let's check the company's underlying financial health.

Copy

```
# Run Fundamentals Analyst
print("\nRunning Fundamentals Analyst...")
fundamentals_analyst_result = run_analyst(fundamentals_analyst_node, initial_st
initial_state['fundamentals_report'] = fundamentals_analyst_result.get('fundame
console.print("----- Fundamentals Analyst Report -----")
console.print(Markdown(initial_state['fundamentals_report']))
Running Fundamentals Analyst...
----- Fundamentals Analyst Report -----
The fundamental picture for NVDA is exceptionally strong, though accompanied by

| Metric              | Status      | Insight
|---------------------|-------------|----------------------------------------
| Revenue Growth      | Exceptional | Data center segment is experiencing hyper
```

| Balance Sheet | Strong | Ample cash reserves provide flexibility |

The final report from the Fundamentals Analyst confirms the strong growth story but also introduces the first note of caution ...

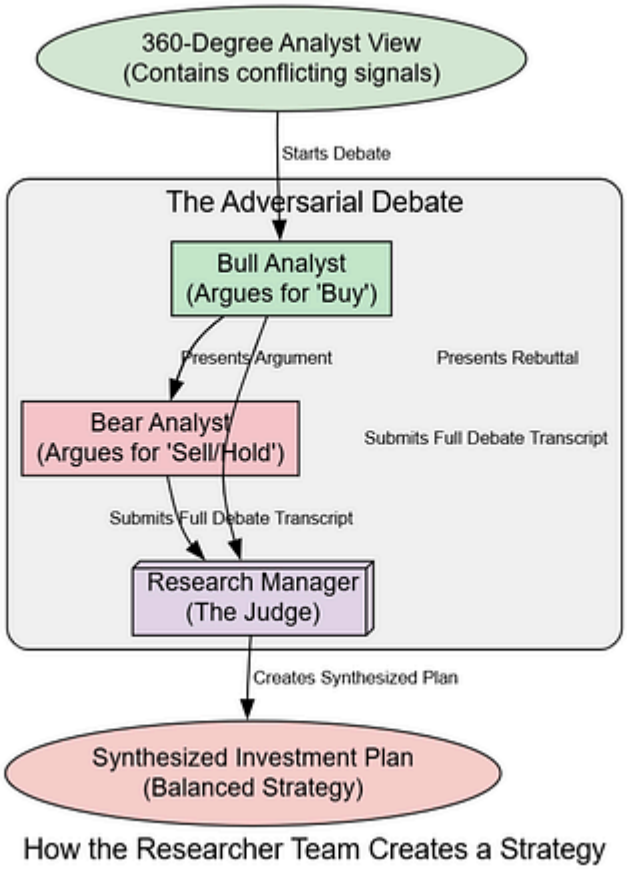
> "premium valuation" and a "High" P/E ratio.

This is a crucial piece of conflicting information. While everything else appears bullish, the high valuation presents a clear risk.

Our `initial_state` object is now fully populated with a rich, multi-faceted view of NVDA's market position. With this comprehensive but now clearly conflicting data, the stage is set for our Researcher Team to debate its meaning and forge a coherent strategy.

Building The Bull vs Bear Researcher Team

With the four analyst reports compiled, our `AgentState` is now packed with raw intelligence. However, raw data can be conflicting and requires interpretation. As we saw, the Fundamentals Analyst introduced a key risk, high valuation that contradicts the otherwise overwhelmingly positive signals.



Resolving Conflicts (Created by Fareed Khan)

This team's purpose is to critically evaluate the evidence by staging a structured, adversarial debate between two opposing viewpoints a **Bull** and a **Bear.** This process is designed to prevent confirmation bias and stress-test the investment

plan.

First, we need to define the logic for these debaters. We'll create a factory function, similar to the one for our analysts, to produce the researcher nodes.

Copy

```
# This function is a factory that creates a LangGraph node for a researcher age
def create_researcher_node(llm, memory, role_prompt, agent_name):
    """
    Creates a node for a researcher agent.
Args:
        llm: The language model instance to be used by the agent.
        memory: The long-term memory instance for this agent to learn from past
        role_prompt: The specific system prompt defining the agent's persona (E
        agent_name: The name of the agent, used for logging and identifying arg
    """
    def researcher_node(state):
        # First, combine all analyst reports into a single summary for context.
        situation_summary = f"""
        Market Report: {state['market_report']}
        Sentiment Report: {state['sentiment_report']}
        News Report: {state['news_report']}
        Fundamentals Report: {state['fundamentals_report']}
        """
        # Retrieve relevant memories from past, similar situations.
        past_memories = memory.get_memories(situation_summary)
```

```
        # Construct the full prompt for the LLM.
        prompt = f"""{role_prompt}
        Here is the current state of the analysis:
        {situation_summary}
        Conversation history: {state['investment_debate_state']['history']}
        Your opponent's last argument: {state['investment_debate_state']['curre
        Reflections from similar past situations: {past_memory_str or 'No past
        Based on all this information, present your argument conversationally.'

        # Invoke the LLM to generate the argument.
        response = llm.invoke(prompt)
        argument = f"{agent_name}: {response.content}"

        # Update the debate state with the new argument.
        debate_state = state['investment_debate_state'].copy()
        debate_state['history'] += "\n" + argument
        # Update the specific history for this agent (Bull or Bear).
        if agent_name == 'Bull Analyst':
            debate_state['bull_history'] += "\n" + argument
        else:
            debate_state['bear_history'] += "\n" + argument
        debate_state['current_response'] = argument
        debate_state['count'] += 1
        return {"investment_debate_state": debate_state}
    return researcher_node
```

The `create_researcher_node` function is the core of our debate logic. Let's look at its key components:

- `past_memories` : Before forming an argument, the agent queries its `memory` object. This is a critical step for learning; it allows the agent to recall lessons from previous trades to inform its current stance.
- Prompt Structure: The prompt is carefully constructed to include the agent's role, the analyst reports, the full debate history, the opponent's most recent argument, and its own long-term memories. This rich context enables sophisticated rebuttals.
- State Updates: The function returns an updated `investment_debate_state` , appending the new argument to the `history` and updating the `count` .

Now, let's define the specific personas for our Bull and Bear and create their nodes.

Copy

```
# The Bull's persona is optimistic, focusing on strengths and growth.
bull_prompt = "You are a Bull Analyst. Your goal is to argue for investing in t

# The Bear's persona is pessimistic, focusing on risks and weaknesses.
bear_prompt = "You are a Bear Analyst. Your goal is to argue against investing
# Create the callable nodes using our factory function.
```

With the debaters ready, we need a judge. The **Research Manager** agent will review the entire debate transcript and produce the final, synthesized investment plan. This is a high-stakes reasoning task, so we will use our powerful `deep_thinking_llm`.

Copy

```
# This function creates the Research Manager node.
def create_research_manager(llm, memory):
    def research_manager_node(state):
        # The prompt instructs the manager to act as a judge and synthesizer.
        prompt = f"""As the Research Manager, your role is to critically evalua
        Summarize the key points, then provide a clear recommendation: Buy, Se

        Debate History:
        {state['investment_debate_state']['history']}"""
        response = llm.invoke(prompt)
        # The output is the final investment plan, which will be passed to the
        return {"investment_plan": response.content}
    return research_manager_node

# Create the callable manager node.
research_manager_node = create_research_manager(deep_thinking_llm, invest_judge
print("Researcher and Manager agent creation functions are now available.")
```

then the Bear will rebut it, with the state being updated after each turn.

Copy

```
# We'll use the state from the end of the Analyst section.
current_state = initial_state

# Loop through the number of debate rounds defined in our config.
for i in range(config['max_debate_rounds']):
    print(f"--- Investment Debate Round {i+1} ---")
    # The Bull goes first.
    bull_result = bull_researcher_node(current_state)
    current_state['investment_debate_state'] = bull_result['investment_debate_s
    console.print("\n**Bull's Argument:**")
    # We parse the response to print only the new argument.
    console.print(Markdown(current_state['investment_debate_state']['current_re
    # Then, the Bear rebuts.
    bear_result = bear_researcher_node(current_state)
    current_state['investment_debate_state'] = bear_result['investment_debate_s
    console.print("\n**Bear's Rebuttal:**")
    console.print(Markdown(current_state['investment_debate_state']['current_re
    print("\n")
# After the loops, store the final debate state back into the main initial_stat
initial_state['investment_debate_state'] = current_state['investment_debate_sta
--- Investment Debate Round 1 ---

**Bulls Argument:**
The case for NVDA is ironclad. We have a perfect alignment across all vectors:
```

```
My opponent sees a perfect picture, but I see a stock priced for a future that

--- Investment Debate Round 2 ---
**Bull Argument:**
The Bear cyclicality argument is outdated. The AI revolution is not a cycle; it
**Bear Rebuttal:**
Calling the AI boom non-cyclical is pure speculation. All industries, especial
```

The output shows our debate working perfectly.

- In Round 1, the Bull presents a strong opening statement based on the confluence of positive data. The Bear immediately counters by seizing on the one negative signal from the analyst reports: the high valuation.
- In Round 2, the debaters engage in direct rebuttals. The Bull dismisses the risk by reframing it ("structural shift"), while the Bear doubles down on it by quantifying the potential downside ("30–40% drawdown").

This debate has successfully surfaced the core tension of the investment case: powerful momentum vs. significant valuation risk. Now, the full transcript is ready for the Research Manager to synthesize into a final plan.

Copy

```
manager_result = research_manager_node(initial_state)
# The manager's output is stored in the 'investment_plan' field.
initial_state['investment_plan'] = manager_result['investment_plan']

console.print("\n----- Research Manager's Investment Plan -----")
console.print(Markdown(initial_state['investment_plan']))
----- Research Manager's Investment Plan -----
After reviewing the spirited debate, the Bull's core argument-that NVDA is a ge
**Recommendation: Buy**
**Rationale:** The confluence of exceptional fundamentals, strong technical mor
**Strategic Actions:** I propose a scaled-entry approach to manage the risks hi
```

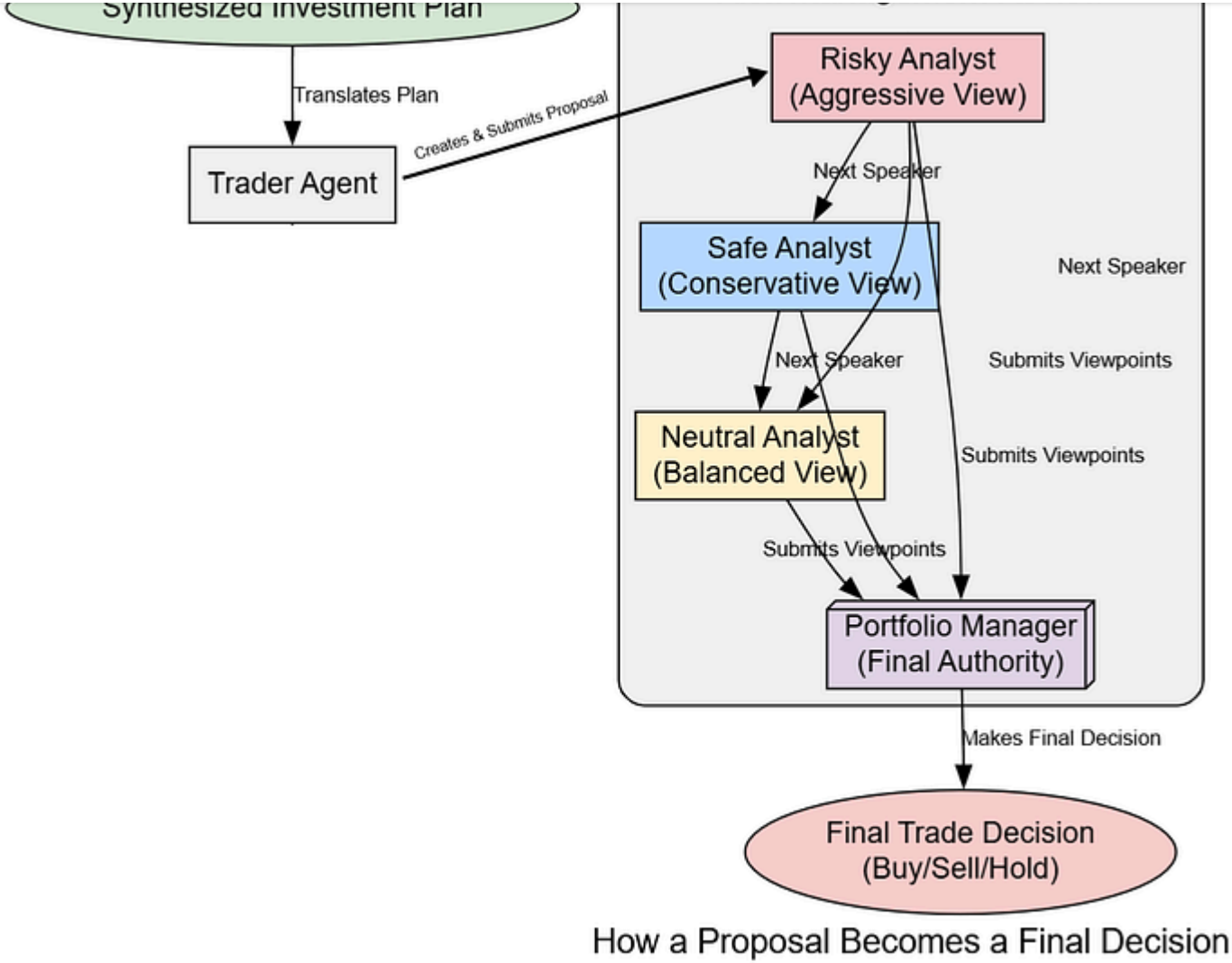
Our Research Manager isn't just picking a side here it creates a new strategy. It endorses the Bull's **"Buy"** recommendation but explicitly incorporates the Bear's concerns by proposing "Strategic Actions" like a scaled entry and a defined stop-loss. This nuanced plan is far more practical and risk-aware than either debater's individual stance.

With a clear investment plan from the research team, the workflow now moves to the execution-focused agents.

Creating The Trader and Risk Management Agents

well-reasoned, is still a high-level document. It needs to be translated into a concrete trading proposal that can be implemented in the market.

This is the job of the **Trader Agent**. Once the Trader formulates a proposal, it's passed to the **Risk Management Team** for final scrutiny. Here, agents with different risk appetites aggressive, conservative, and neutral, debate the plan to ensure all angles are considered before capital is committed.



Trade and RM (Created by Fareed Khan)

is that its response must end with a specific, machine-readable tag.

Copy

```
import functools

# This function creates the Trader agent node.
def create_trader(llm, memory):
    def trader_node(state, name):
        # The prompt is simple: take the plan and create a proposal.
        # The key instruction is the mandatory final tag.
        prompt = f"""You are a trading agent. Based on the provided investment
        Your response must end with 'FINAL TRANSACTION PROPOSAL: **BUY/HOLD/SEL

        Proposed Investment Plan: {state['investment_plan']}"""
        result = llm.invoke(prompt)
        # The output updates the state with the trader's plan and identifies th
        return {"trader_investment_plan": result.content, "sender": name}
    return trader_node
```

The design of this Trader agent is focused on clarity and actionability. The prompt's most critical part is the instruction to end with `FINAL TRANSACTION PROPOSAL: **BUY/HOLD/SELL**`. This isn't just for human readability; it creates a predictable pattern that downstream processes (like our signal extractor later on) can reliably parse.

personas. The logic here is more complex, as each agent needs to be aware of its opponents' most recent arguments to formulate a proper rebuttal.

Copy

```
# This function is a factory for creating the risk debater nodes.
def create_risk_debator(llm, role_prompt, agent_name):
    def risk_debator_node(state):
        # First, get the arguments from the other two debaters from the state.
        risk_state = state['risk_debate_state']
        opponents_args = []
        if agent_name != 'Risky Analyst' and risk_state['current_risky_response
        if agent_name != 'Safe Analyst' and risk_state['current_safe_response']
        if agent_name != 'Neutral Analyst' and risk_state['current_neutral_resp

        # Construct the prompt with the trader's plan, debate history, and oppo
        prompt = f"""{role_prompt}
        Here is the trader's plan: {state['trader_investment_plan']}
        Debate history: {risk_state['history']}
        Your opponents' last arguments:\n{'\n'.join(opponents_args)}
        Critique or support the plan from your perspective."""

        response = llm.invoke(prompt).content

        # Update the risk debate state with the new argument.
        new_risk_state = risk_state.copy()
        new_risk_state['history'] += f"\n{agent_name}: {response}"
```

```
        elif agent_name == 'Safe Analyst': new_risk_state['current_safe_respons
        else: new_risk_state['current_neutral_response'] = response
        new_risk_state['count'] += 1
        return {"risk_debate_state": new_risk_state}
    return risk_debator_node
```

The logic in `create_risk_debator` is what enables a true multi-party debate. By dynamically building the `opponents_args` list, we ensure that each agent is directly responding to the other participants, not just stating its opinion in isolation.

The state update is also more granular, populating fields like `current_risky_response` so that the other agents can access them in the next turn.

Now, we'll define the specific personas for our three risk agents. This adversarial setup, one aggressive, one conservative, one balanced is designed to stress-test the Trader's plan from all angles.

Copy

```
# The Risky persona advocates for maximizing returns, even if it means higher r
risky_prompt = "You are the Risky Risk Analyst. You advocate for high-reward op
```

```
# The Neutral persona provides a balanced, objective viewpoint.
neutral_prompt = "You are the Neutral Risk Analyst. You provide a balanced pers
```

With the agent logic defined, we can now instantiate all the callable nodes for this part of the workflow.

```
Copy

# Create the Trader node. We use functools.partial to pre-fill the 'name' argu
trader_node_func = create_trader(quick_thinking_llm, trader_memory)
trader_node = functools.partial(trader_node_func, name="Trader")

# Create the three risk debater nodes using their specific prompts.
risky_node = create_risk_debator(quick_thinking_llm, risky_prompt, "Risky Analy
safe_node = create_risk_debator(quick_thinking_llm, safe_prompt, "Safe Analyst"
neutral_node = create_risk_debator(quick_thinking_llm, neutral_prompt, "Neutra
```

Now let's run the Trader agent on the `investment_plan` we generated in the previous section.

```
Copy

print("Running Trader...")
# Pass the current state to the trader node.
trader_result = trader_node(initial_state)
```

```
console.print("\n----- Trader's Proposal -----")
console.print(Markdown(initial_state['trader_investment_plan']))
Running Trader...

----- Trader's Proposal -----
The Research Manager's plan to scale into a long position is prudent and well-s
I will execute this by establishing an initial 50% position at the market open.
FINAL TRANSACTION PROPOSAL: **BUY**
```

The Trader's output is excellent is showing that it has successfully translated the Research Manager' strategic guidance into a concrete, actionable plan with specific parameters (50% position, entry/exit points). Crucially, it has also included the `FINAL TRANSACTION PROPOSAL` tag, making its core recommendation unambiguous.

This proposal now goes to the Risk Management team for their debate.

```
                                    Copy

print("--- Risk Management Debate Round 1 ---")

risk_state = initial_state
# We run the debate for the number of rounds specified in our config (currently
for _ in range(config['max_risk_discuss_rounds']):
```

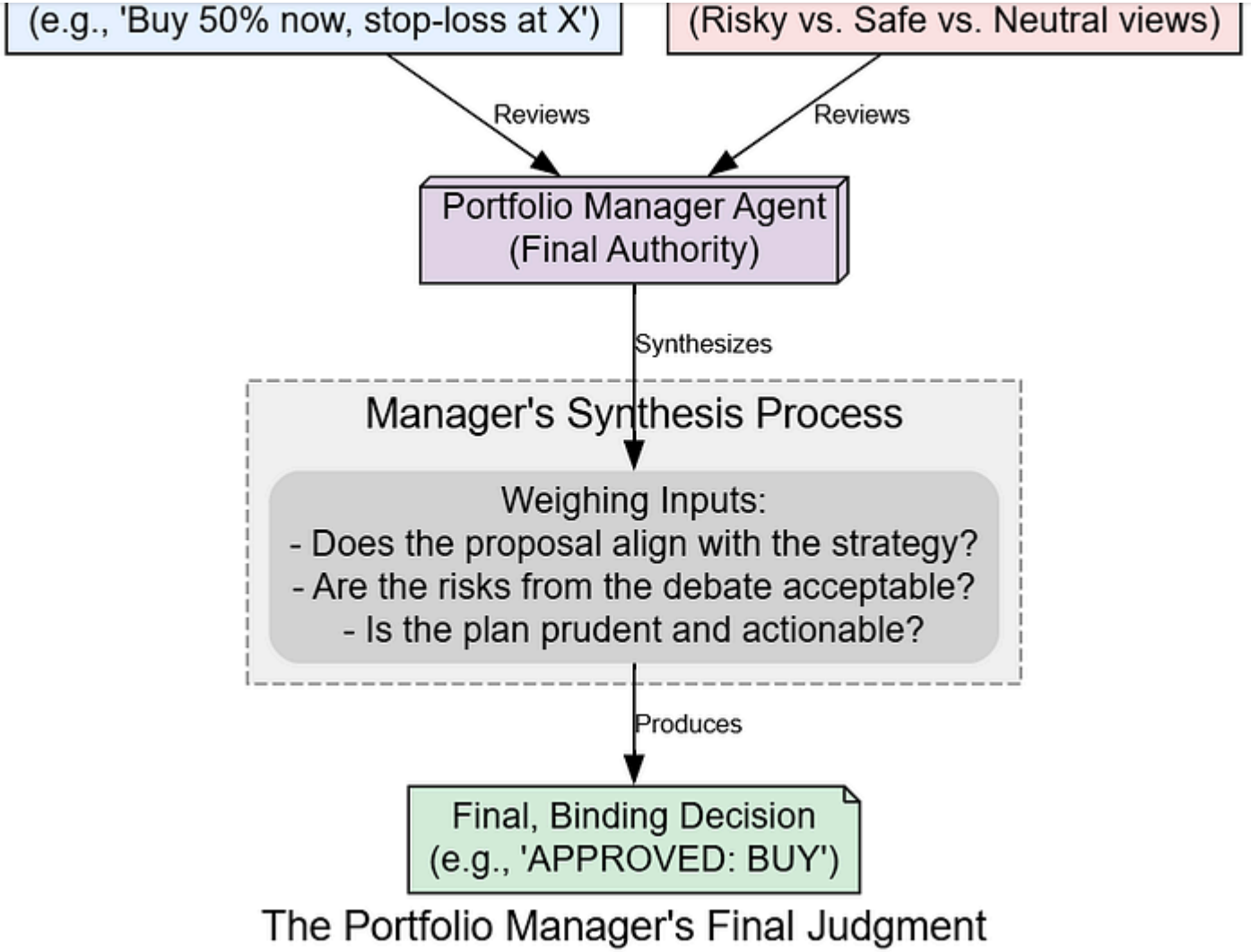
```
    risky_result = risky_node(risk_state)
    risk_state['risk_debate_state'] = risky_result['risk_debate_state']
    console.print("\n**Risky Analyst's View:**")
    console.print(Markdown(risk_state['risk_debate_state']['current_risky_respo

    # Then the Safe analyst.
    safe_result = safe_node(risk_state)
    risk_state['risk_debate_state'] = safe_result['risk_debate_state']
    console.print("\n**Safe Analyst's View:**")
    console.print(Markdown(risk_state['risk_debate_state']['current_safe_respor

    # Finally, the Neutral analyst.
    neutral_result = neutral_node(risk_state)
    risk_state['risk_debate_state'] = neutral_result['risk_debate_state']
    console.print("\n**Neutral Analyst's View:**")
    console.print(Markdown(risk_state['risk_debate_state']['current_neutral_res

# Update the main state with the final debate transcript.
initial_state['risk_debate_state'] = risk_state['risk_debate_state']
--- Risk Management Debate Round 1 ---
**Risky Analyst's View:**
The scaled entry plan is too conservative. All data points to immediate and cor
**Safe Analyst's View:**
A full position would be irresponsible. The stock is trading at a high valuatic
**Neutral Analyst's View:**
The trader's plan is excellent and requires no modification. It perfectly balar
```

position"), the Safe Analyst pushes for tighter controls ("tighter stop-loss"), and the Neutral Analyst validates the Trader's plan as a well-balanced compromise. This debate has effectively illuminated the full spectrum of risk considerations.

Portfolio Manager Binding Decision

The final step in the decision-making process rests with the **Portfolio Manager** agent.

> This agent acts as the head of the firm. It reviews the Trader's plan and the entire risk debate, then issues the final, binding decision.



Judgment Process (creaed by Fareed Khan)

First, let's define the agent function.

Copy

```
# This function creates the Portfolio Manager node.
def create_risk_manager(llm, memory):
    def risk_manager_node(state):
        # The prompt asks for a final, binding decision based on all prior work
        prompt = f"""As the Portfolio Manager, your decision is final. Review t
        Provide a final, binding decision: Buy, Sell, or Hold, and a brief just

        Trader's Plan: {state['trader_investment_plan']}
        Risk Debate: {state['risk_debate_state']['history']}"""
        response = llm.invoke(prompt).content
        # The output is stored in the 'final_trade_decision' field of the state
        return {"final_trade_decision": response}
    return risk_manager_node

# Create the callable manager node.
risk_manager_node = create_risk_manager(deep_thinking_llm, risk_manager_memory)
```

Now, let's run this final agent to get our decision.

Copy

```
risk_manager_result = risk_manager_node(initial_state)
# The manager's output is stored in the 'final_trade_decision' field.
initial_state['final_trade_decision'] = risk_manager_result['final_trade_decisi

console.print("\n----- Portfolio Manager's Final Decision -----")
console.print(Markdown(initial_state['final_trade_decision']))
Running Portfolio Manager for final decision...

----- Portfolio Manager's Final Decision -----
Having reviewed the trader's well-reasoned plan and the comprehensive risk deba
The plan is sound and aligns with our firm's goal of capturing growth while mar
**Final Decision: BUY**
```

The final output shows the Portfolio Manager synthesizing all the prior stages. It explicitly references the risk debate, agrees with the Neutral Analyst, and approves the Trader's plan. This confirms the **BUY** decision and concludes the core analytical workflow. We have successfully transformed raw, multi-source data into a single, reasoned, and approved trading decision.

Now that all the individual agent components have been defined and tested, we are ready to assemble them into a single, automated workflow using LangGraph.

Orchestrating the Agent Society

However, they currently exist as isolated functions. The final step in construction is to assemble them into a cohesive, automated workflow using LangGraph's `StateGraph`.

This involves programmatically defining the "nervous system" of our agent society. We will wire the nodes together with edges and, most importantly, define the conditional logic that will route the `AgentState` through the correct sequence of agents, manage the debate loops, and handle the ReAct tool-use cycles.

First, we need to create the helper functions that will act as the "traffic controllers" for our graph.

Copy

```
from langchain_core.messages import HumanMessage, RemoveMessage
from langgraph.prebuilt import tools_condition

# The ConditionalLogic class holds the routing functions for our graph.
class ConditionalLogic:
    def __init__(self, max_debate_rounds=1, max_risk_discuss_rounds=1):
        # Store the maximum number of rounds from our config.
        self.max_debate_rounds = max_debate_rounds
        self.max_risk_discuss_rounds = max_risk_discuss_rounds
```

```
        # The tools_condition helper checks the last message in the state.
        # If it's a tool call, it returns "tools". Otherwise, it returns "conti
        return "tools" if tools_condition(state) == "tools" else "continue"

    # This function controls the flow of the investment debate.
    def should_continue_debate(self, state: AgentState) -> str:
        # If the debate has reached its maximum rounds, route to the Research M
        if state["investment_debate_state"]["count"] >= 2 * self.max_debate_rou
            return "Research Manager"
        # Otherwise, continue the debate by alternating speakers.
        return "Bear Researcher" if state["investment_debate_state"]["current_r

    # This function controls the flow of the risk management discussion.
    def should_continue_risk_analysis(self, state: AgentState) -> str:
        # If the risk discussion has reached its maximum rounds, route to the j
        if state["risk_debate_state"]["count"] >= 3 * self.max_risk_discuss_rou
            return "Risk Judge"
        # Otherwise, continue the discussion by cycling through speakers.
        speaker = state["risk_debate_state"]["latest_speaker"]
        if speaker == "Risky Analyst": return "Safe Analyst"
        if speaker == "Safe Analyst": return "Neutral Analyst"
        return "Risky Analyst"

# Instantiate the logic class with values from our central config.
conditional_logic = ConditionalLogic(
    max_debate_rounds=config['max_debate_rounds'],
    max_risk_discuss_rounds=config['max_risk_discuss_rounds']
)
```

- `should_continue_analyst` : This is the core of the ReAct loop. It inspects the agent's output and decides whether to route to the `tools` node for execution or to `continue` to the next agent.
- `should_continue_debate` & `should_continue_risk_analysis` : These methods control the debate loops. They check the `count` in their respective sub-states. If the count exceeds the limit from our `config` , they break the loop and route to the manager/judge. Otherwise, they route to the next debater in the sequence.

> A key challenge in multi-agent systems is preventing context from one task from "leaking" into another and confusing the LLM.

After an analyst completes its multi-step ReAct loop, its `messages` state will be cluttered with tool calls and intermediate reasoning. We need a way to clean this up before the next analyst begins.

Copy

```
# This function creates a node that clears the messages from the state.
def create_msg_delete():
    # This helper function is designed to be used as a node in the graph.
    def delete_messages(state):
```

```
        return {"messages": [RemoveMessage(id=m.id) for m in state["messages"]]}
    return delete_messages

# Create the callable message clearing node.
msg_clear_node = create_msg_delete()
```

The `msg_clear_node` is a simple but crucial utility. By placing this node between our analyst agents, we ensure that each analyst starts with a clean slate, receiving only the global state information (like the reports) without the conversational clutter from the previous agent's tool usage.

Next, we need a dedicated node to *execute* the tool calls that the analysts generate.

Copy

```
from langgraph.prebuilt import ToolNode

# Create a list of all tools available in our toolkit.
all_tools = [
    toolkit.get_yfinance_data,
    toolkit.get_technical_indicators,
    toolkit.get_finnhub_news,
    toolkit.get_social_media_sentiment,
    toolkit.get_fundamental_analysis,
    toolkit.get_macroeconomic_news
]
```

```
tool_node = ToolNode(all_tools)
```

The analyst agents decide to call a tool, but this `tool_node` is what actually *runs* the corresponding Python function. This separation of concerns is a core LangGraph pattern. A single `tool_node` can serve all four of our analyst agents, making the graph efficient.

Now for the main event. We will create a `StateGraph` instance and programmatically add all our agent nodes, tool nodes, and the edges that connect them. This code will translate our conceptual workflow into a concrete, executable graph object.

Copy

```
from langgraph.graph import StateGraph, START, END

# Initialize a new StateGraph with our main AgentState.
workflow = StateGraph(AgentState)

# --- Add all nodes to the graph ---
# Add Analyst Team Nodes
workflow.add_node("Market Analyst", market_analyst_node)
workflow.add_node("Social Analyst", social_analyst_node)
workflow.add_node("News Analyst", news_analyst_node)
workflow.add_node("Fundamentals Analyst", fundamentals_analyst_node)
```

```
# Add Researcher Team Nodes
workflow.add_node("Bull Researcher", bull_researcher_node)
workflow.add_node("Bear Researcher", bear_researcher_node)
workflow.add_node("Research Manager", research_manager_node)

# Add Trader and Risk Team Nodes
workflow.add_node("Trader", trader_node)
workflow.add_node("Risky Analyst", risky_node)
workflow.add_node("Safe Analyst", safe_node)
workflow.add_node("Neutral Analyst", neutral_node)
workflow.add_node("Risk Judge", risk_manager_node)

# --- Wire the nodes together with edges ---
# Set the entry point for the entire workflow.
workflow.set_entry_point("Market Analyst")

# Define the sequential flow and ReAct loops for the Analyst Team.
workflow.add_conditional_edges("Market Analyst", conditional_logic.should_cont
workflow.add_edge("tools", "Market Analyst") # After a tool call, loop back to
workflow.add_edge("Msg Clear", "Social Analyst")
workflow.add_conditional_edges("Social Analyst", conditional_logic.should_cont
workflow.add_edge("tools", "Social Analyst")
workflow.add_conditional_edges("News Analyst", conditional_logic.should_continu
workflow.add_edge("tools", "News Analyst")
workflow.add_conditional_edges("Fundamentals Analyst", conditional_logic.should
workflow.add_edge("tools", "Fundamentals Analyst")

# Define the research debate loop.
```

```
workflow.add_edge("Research Manager", "Trader")

# Define the risk debate loop.
workflow.add_edge("Trader", "Risky Analyst")
workflow.add_conditional_emails("Risky Analyst", conditional_logic.should_cont
workflow.add_conditional_edges("Safe Analyst", conditional_logic.should_continu
workflow.add_conditional_edges("Neutral Analyst", conditional_logic.should_cont

# Define the final edge to the end of the workflow.
workflow.add_edge("Risk Judge", END)
```

This block is the programmatic definition of our entire workflow diagram. We first use `.add_node()` to register all our agent functions. Then, we use `.add_edge()` for direct transitions (e.g., `Research Manager` always goes to `Trader`) and `.add_conditional_edges()` for dynamic routing. The conditional edges use our `conditional_logic` object to decide the next step, enabling the complex looping behavior for the ReAct cycles and debates. The special `START` and `END` keywords define the graph's entry and exit points.

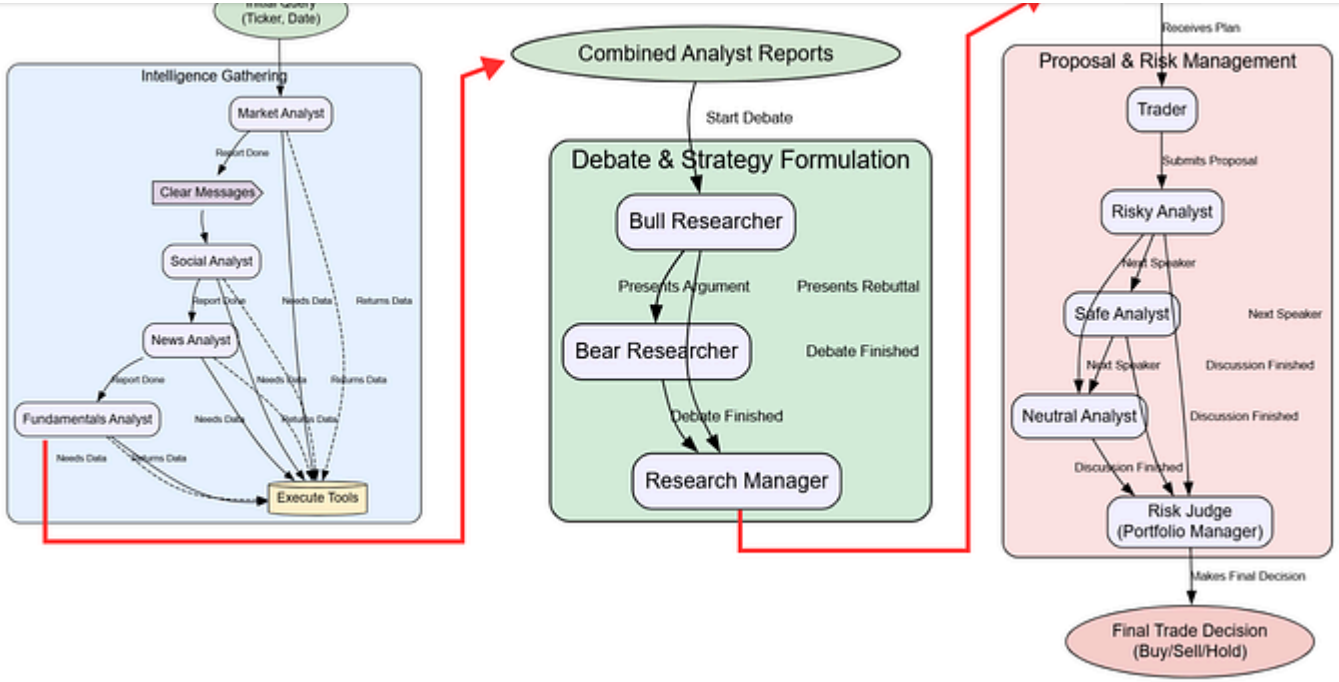
Compiling and Visualizing the Agentic Workflow

The graph has been defined, but it's still just a blueprint. To make it executable, we need to `.compile()` it. This step takes our definition and creates a highly

A major advantage of using LangGraph is its built-in visualization capability. Visualizing the graph is an incredibly useful step to verify that our complex wiring is correct before we run it.

Copy

```
# The compile() method finalizes the graph and makes it ready for execution.
trading_graph = workflow.compile()
print("Graph compiled successfully.")

# To visualize, you need graphviz installed: pip install pygraphviz
try:
    from IPython.display import Image, display
    # The get_graph() method returns a representation of the graph structure.
    # The draw_png() method renders this structure as a PNG image.
    png_image = trading_graph.get_graph().draw_png()
    display(Image(png_image))
except Exception as e:
    print(f"Graph visualization failed: {e}. Please ensure pygraphviz is instal
```



Workflow Plot (Created by Fareed Khan)

The output is a visual confirmation of our entire agentic system. We can clearly see the intended flow:

- The initial sequential chain of four **Analyst** nodes, each with its own ReAct loop that routes through the central **tools** node.
- The transition to the **Researcher** debate, with the conditional edges creating a loop between the **Bull** and **Bear** nodes before proceeding to the **Research Manager**.

- The final step where the **Risk Judge** (our Portfolio Manager) concludes the process, leading to **END**.

This visual verification gives us confidence that our orchestration is correct. With the compiled `trading_graph` object ready, we can now proceed to the final execution.

Executing the End-to-End Trading Analysis

All our components are built, the graph is wired, and the `trading_graph` object is compiled and ready. We can now invoke the entire multi-agent system with a single command. We will provide it with our target ticker and date, and then stream the results to watch the agents collaborate in real-time as they move through the complex workflow we've designed.

First, let's prepare the initial input state, which is the starting point for the entire graph execution.

```
Copy
```

```
    messages=[HumanMessage(content=f"Analyze {TICKER} for trading on {TRADE_DAT
    company_of_interest=TICKER,
    trade_date=TRADE_DATE,
    # Initialize the debate states with default empty values to ensure a clean
    investment_debate_state=InvestDebateState({'history': '', 'current_response
    risk_debate_state=RiskDebateState({'history': '', 'latest_speaker': '','cu
)

print(f"Running full analysis for {TICKER} on {TRADE_DATE}")


#### OUTPUT ###
Running full analysis for NVDA on 2025-9-4
```

We'll use the `.stream()` method to invoke the graph. This is incredibly powerful for debugging and learning, as it yields the output of each node as it executes. We'll print the name of the node as it runs to trace the workflow and see our orchestrated system in action.

Copy

```
final_state = None
print("\n--- Invoking Graph Stream ---")
# Set the recursion limit from our config, a safety measure for complex graphs.
graph_config = {"recursion_limit": config['max_recur_limit']}
```

```
    # The "chunk" is a dictionary where the key is the name of the node that ju
    node_name = list(chunk.keys())[0]
    print(f"Executing Node: {node_name}")
    # We keep track of the final state to analyze it after the run.
    final_state = chunk[node_name]
print("\n--- Graph Stream Finished ---")
```

Once we run this code our entire system will start acting rightaway, take a look at the output.

Copy

```
--- Invoking Graph Stream ---
Executing Node: Market Analyst
Executing Node: tools
Executing Node: Market Analyst
Executing Node: Msg Clear
Executing Node: Social Analyst
Executing Node: tools
Executing Node: Social Analyst
Executing Node: News Analyst
Executing Node: tools
Executing Node: News Analyst
Executing Node: Fundamentals Analyst
Executing Node: tools
Executing Node: Fundamentals Analyst
Executing Node: Bull Researcher
```

```
Executing Node: Bear Researcher
Executing Node: Research Manager
Executing Node: Trader
Executing Node: Risky Analyst
Executing Node: Safe Analyst
Executing Node: Neutral Analyst
Executing Node: Risk Judge

--- Graph Stream Finished ---
```

The output trace provides a perfect, real-time log of our graph's execution path. We can see the ReAct loops in action for the analysts (e.g., `Market Analyst -> tools -> Market Analyst`), the `Msg Clear` node firing between them, the back-and-forth of the research debate, and the final sequence through the trader and risk teams. This confirms that our conditional logic and wiring are functioning exactly as intended.

The stream has finished, and the complete, enriched `AgentState` is now stored in our `final_state` variable. Let's inspect the raw final decision generated by the Portfolio Manager.

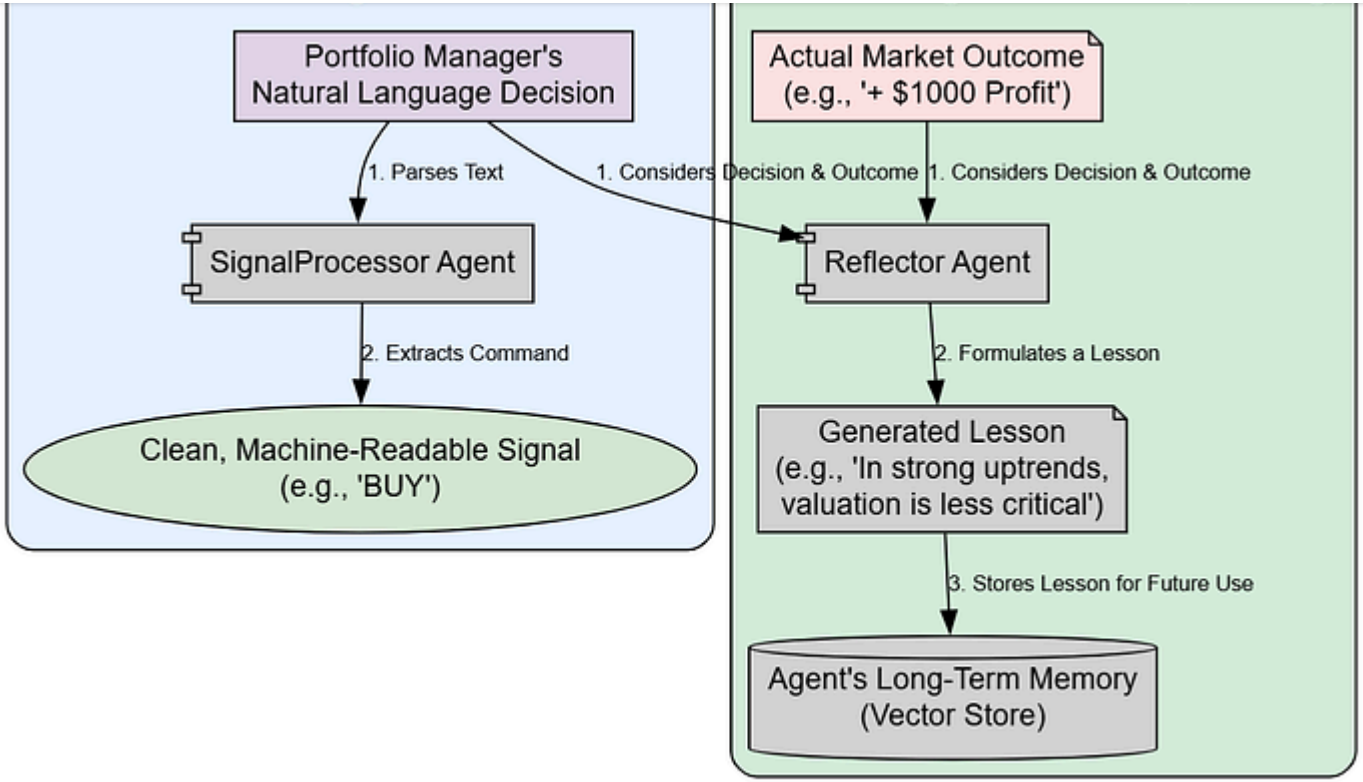
Copy

```
####### OUTPUT ############
----- Final Raw Output from Portfolio Manager -----
Having reviewed the trader's well-reasoned plan and the comprehensive risk deba

The plan is sound and aligns with our firm's goal of capturing growth while mar
**Final Decision: BUY**
```

The final output is consistent with our manual tests, confirming that the fully automated graph can replicate the logical flow we established earlier. The system has successfully produced a detailed, reasoned decision.

Extracting Clean Signals and Agent Reflection

Our pipeline has successfully produced a decision. However, for practical use, we need two final pieces:



Final Steps: Signal Extraction & Agent Reflection

Signal and Reflection (Created by Fareed Khan)

1. **A clean, machine-readable final signal** (BUY, SELL, or HOLD). The Portfolio Manager's output is natural language; we need to distill it into a single, unambiguous command.
2. **A mechanism for the agents to learn from the outcome.** This is what turns a one-off analysis into a system that can improve over time.

Copy

```
class SignalProcessor:
    # This class is responsible for parsing the final LLM output into a clean,
    def __init__(self, llm):
        self.llm = llm

def process_signal(self, full_signal: str) -> str:
        # We use a simple, focused prompt to ask the LLM for the single-word de
        messages = [
            ("system", "You are an assistant designed to extract the final inve
            ("human", full_signal),
        ]
        result = self.llm.invoke(messages).content.strip().upper()
        # Basic validation to ensure the output is one of the three expected si
        if result in ["BUY", "SELL", "HOLD"]:
            return result
        return "ERROR_UNPARSABLE_SIGNAL"
# Instantiate the processor with our quick_thinking_llm.
signal_processor = SignalProcessor(quick_thinking_llm)
final_signal = signal_processor.process_signal(final_state['final_trade_decisio
print(f"Extracted Signal: {final_signal}")
####### OUTPUT #######
Extracted Signal: BUY
```

analyze their performance and formulate a concise lesson, which is then stored in their long-term `FinancialSituationMemory`.

Copy

```
class Reflector:
    # This class orchestrates the learning process for the agents.
    def __init__(self, llm):
        self.llm = llm
        # This prompt guides the agent to reflect on its performance.
        self.reflection_prompt = """You are an expert financial analyst. Review
        - First, determine if the decision was correct or incorrect based on th
        - Analyze the most critical factors that led to the success or failure.
        - Finally, formulate a concise, one-sentence lesson or heuristic that c

        Market Context & Analysis: {situation}
        Outcome (Profit/Loss): {returns_losses}"""

def reflect(self, current_state, returns_losses, memory, component_key_func):
        # The component_key_func is a lambda function to extract the specific t
        situation = f"Reports: {current_state['market_report']} {current_state
        prompt = self.reflection_prompt.format(situation=situation, returns_los
        result = self.llm.invoke(prompt).content
        # The situation (context) and the generated lesson (result) are stored
        memory.add_situations([(situation, result)])
print("SignalProcessor and Reflector classes defined.")
```

our learning agents.

```
Copy

print("Simulating reflection based on a hypothetical profit of $1000...")

reflector = Reflector(quick_thinking_llm)
hypothetical_returns = 1000

# Run the reflection process for each agent with memory.
print("Reflecting and updating memory for Bull Researcher...")
reflector.reflect(final_state, hypothetical_returns, bull_memory, lambda s: s[

print("Reflecting and updating memory for Bear Researcher...")
reflector.reflect(final_state, hypothetical_returns, bear_memory, lambda s: s[

print("Reflecting and updating memory for Trader...")
reflector.reflect(final_state, hypothetical_returns, trader_memory, lambda s: s

print("Reflecting and updating memory for Risk Manager...")
reflector.reflect(final_state, hypothetical_returns, risk_manager_memory, lambd
```

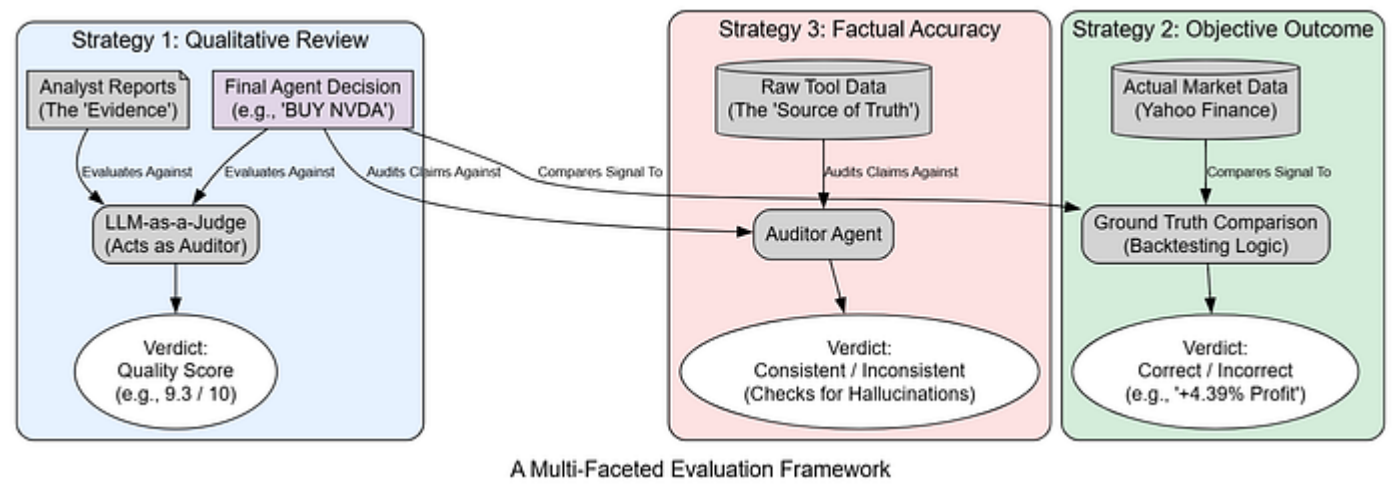
The reflection process is now complete. Each agent has analyzed the successful `BUY` decision in the context of the market reports and the hypothetical profit.

their respective vector memories. The next time a similar situation arises, they will retrieve these lessons, making the entire system "smarter."

Auditing the System using 3 Evaluation Strategies

While our pipeline produces a decision, how can we be sure it's a *good* one?

> A production system requires automated ways to score the quality of its output.



Evaluation Strategies (Created by Fareed Khan)

and factual accuracy.

First, we'll use a powerful LLM agent as an impartial **LLM-as-a-Judge**, scoring the final decision on key criteria.

Copy

```
from pydantic import BaseModel, Field
from langchain_core.prompts import ChatPromptTemplate

# Define a structured schema for evaluation results.
# This ensures the LLM outputs scores in a controlled format.
class Evaluation(BaseModel):
    reasoning_quality: int = Field(description="Score 1-10 on the coherence and
    evidence_based_score: int = Field(description="Score 1-10 on citation of ev
    actionability_score: int = Field(description="Score 1-10 on how clear and a
    justification: str = Field(description="A brief justification for the score

# Create a prompt template for the evaluator model.
# The prompt instructs the LLM to act like a financial auditor.
evaluator_prompt = ChatPromptTemplate.from_template(
    """You are an expert financial auditor. Evaluate the 'Final Trading Decisio
    Analyst Reports:
    {reports}
    Final Trading Decision to Evaluate:
    {final_decision}
    """
```

```
# Combine the prompt with an LLM, enforcing structured output via the Evaluati
# `deep_thinking_llm` here is assumed to be a previously defined LLM instance.
evaluator_chain = evaluator_prompt | deep_thinking_llm.with_structured_output(E

# Build the full text summary of analyst reports from final_state.
reports_summary = (
    f"Market: {final_state['market_report']}\n"
    f"Sentiment: {final_state['sentiment_report']}\n"
    f"News: {final_state['news_report']}\n"
    f"Fundamentals: {final_state['fundamentals_report']}"
)

# Prepare evaluator input with both reports and the decision to evaluate.
eval_input = {
    "reports": reports_summary,
    "final_decision": final_state['final_trade_decision']
}

# Run the evaluator chain — returns structured Evaluation object.
evaluation_result = evaluator_chain.invoke(eval_input)

# Print the evaluation report in a readable format.
print("----- LLM-as-a-Judge Evaluation Report -----")
pprint(evaluation_result.dict())
###### OUTPUT ########
----- LLM-as-a-Judge Evaluation Report -----
{'actionability_score': 10,
 'evidence_based_score': 9,
 'justification': "The final decision demonstrates strong, coherent reasoning.
```

```
                    management plan (scaled entry, stop-loss). The decision is '
                    "highly actionable, providing a clear BUY signal and "
                    'approving a specific execution strategy.',
 'reasoning_quality': 9}
```

The LLM-as-a-Judge gives high scores across the board, validating the qualitative strength of the final decision. The high `evidence_based_score` is particularly important, as it confirms the final decision is well-grounded in the initial analyst reports.

Next, the most objective test: **Ground Truth Comparison.** Did the agent's decision actually make money?

Copy

```
def evaluate_ground_truth(ticker, trade_date, signal):
    try:
        # Parse the input trade date and define an evaluation window of 8 days
        start_date = datetime.strptime(trade_date, "%Y-%m-%d").date()
        end_date = start_date + timedelta(days=8)

        # Download market data from Yahoo Finance
        data = yf.download(
            ticker,
            start=start_date.isoformat(),
```

```
)

# Ensure enough trading days exist (at least 5 for evaluation)
if len(data) < 5:
    return "Insufficient data for ground truth evaluation."

# Find the first trading day index (accounting for weekends/holidays)
first_trading_day_index = 0
while data.index[first_trading_day_index].date() < start_date:
    first_trading_day_index += 1
    if first_trading_day_index >= len(data) - 5:
        return "Could not align trade date."

# Get opening price on the aligned trade date
open_price = data['Open'].iloc[first_trading_day_index]

# Get closing price 5 trading days later
close_price_5_days_later = data['Close'].iloc[first_trading_day_index -

# Compute % change over 5 days
performance = ((close_price_5_days_later - open_price) / open_price) *

# Default evaluation result
result = "INCORRECT DECISION"

# Rule-based correctness:
# - BUY is correct if price went up > +1%
# - SELL is correct if price went down < -1%
# - HOLD is correct if price stayed roughly flat (-1% to +1%)
```

```
            (signal == "HOLD" and -1 <= performance <= 1)):
            result = "CORRECT DECISION"

        # Return a detailed evaluation report
        return (f"----- Ground Truth Evaluation Report -----\n"
                f"Agent Signal: {signal} on {trade_date}\n"
                f"Opening Price on {data.index[first_trading_day_index].strftim
                f"Closing Price 5 days later ({data.index[first_trading_day_inc
                f"Actual Market Performance: {performance:+.2f}%\n"
                f"Evaluation Result: {result}")

    # Catch-all error handling (network issues, bad ticker, etc.)
    except Exception as e:
        return f"Ground truth evaluation failed: {e}"


# Example usage: evaluates if the agent's signal was correct in hindsight
ground_truth_report = evaluate_ground_truth(TICKER, TRADE_DATE, final_signal)
print(ground_truth_report)
###### OUTPUT ##########
----- Ground Truth Evaluation Report -----
Agent Signal: BUY on 2024-10-25
Opening Price on 2024-10-25: $128.50
Closing Price 5 days later (2024-11-01): $134.15
Actual Market Performance: +4.39%
Evaluation Result: CORRECT DECISION
```

validation of the system's performance in this specific instance.

Finally, we'll perform a **Factual Consistency Audit** to check if the agents are hallucinating or misrepresenting data. We'll create an 'Auditor' agent to compare claims made in the Market Analyst's report against data fetched directly from the tool.

Copy

```
from pydantic import BaseModel, Field
from langchain_core.prompts import ChatPromptTemplate

# Define schema for audit results
class Audit(BaseModel):
    is_consistent: bool = Field(description="Whether the report is factually co
    discrepancies: list[str] = Field(description="A list of any identified disc
    justification: str = Field(description="A brief justification for the audit

# Prompt template for the auditing task
# The auditor compares the raw data (truth source) against the agent's report
auditor_prompt = ChatPromptTemplate.from_template(
    """You are an auditor. Compare the 'Agent Report' against the 'Raw Data' ar
    Ignore differences in formatting or summarization, but flag any direct cont
    that are not supported by the data.

    Raw Data:
```

```
    Agent Report to Audit:
    {agent_report}
    """
)

# Chain: prompt → deep_thinking_llm → structured output following the Audit sch
auditor_chain = auditor_prompt | deep_thinking_llm.with_structured_output(Audit

# Pull ~60 days of technical indicator data for context leading up to the trade
start_date_audit = (
    datetime.strptime(TRADE_DATE, "%Y-%m-%d") - timedelta(days=60)
).strftime('%Y-%m-%d')

raw_market_data_for_audit = toolkit.get_technical_indicators(TICKER, start_date

# Build input for the auditor (raw technical data + agent's narrative report)
audit_input = {
    "raw_data": raw_market_data_for_audit,
    "agent_report": final_state['market_report']
}

# Run the chain → structured audit output
audit_result = auditor_chain.invoke(audit_input)

# Pretty-print the audit results
print("----- Factual Consistency Audit Report -----")
pprint(audit_result.dict())
###### OUTPUT #######
----- Factual Consistency Audit Report -----
```

```
    'justification': 'The agent's report is factually consistent with the raw '
                     "data. It correctly identifies the bullish MACD, the upward "
                     'trend of the SMAs, and the expanding volatility shown by the '
                     'Bollinger Bands. There are no hallucinations or '
                     'misrepresentations of the provided technical indicators.'}
```

The audit passes with `is_consistent: True`, confirming that our Market Analyst is not hallucinating and is accurately reporting on the data it retrieves from its tools. This is a crucial check for building trust in the system's outputs.

Key Takeaways and Future Directions

We have successfully built, executed, and evaluated a complex, standalone multi-agent financial analysis pipeline from scratch. By replicating the structure of a real-world trading firm, we've demonstrated how specialized agents can collaborate to transform raw, multi-source live data into a single, reasoned, and profitable trading decision.

Key Takeaways:

- Assigning specific roles to different agents allows for deeper, more focused analysis at each stage.

- `LangGraph` can be enough for creating a deep thinking framework for managing the complex state and conditional logic required for such a system to function automatically.
- A robust evaluation framework combines qualitative checks (LLM-as-a-Judge), objective outcomes (Ground Truth), and process checks (Factual Consistency, Tool Usage).

Future Directions:

- The next logical step is to run this pipeline over thousands of historical data points to statistically evaluate its long-term performance (Sharpe ratio, max drawdown, etc.).
- More sophisticated tools could be added, such as those for analyzing options data, economic calendars, or alternative datasets.
- A more advanced supervisor could dynamically choose which analysts to deploy based on the specific stock or market conditions, optimizing for cost and relevance.

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